

EXPERIMENT – 11 - STUDY OF GPIO & BIT ADDRESSING IN AT89C51

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PROGRAM

```
#include <reg51.h>
```

```
sbit LED1 = P1^0;
```

```
sbit LED2 = P1^1;
```

```
void delay()
```

```
{
```

```
    unsigned int i, j;
```

```
    for(i = 0; i < 200; i++)
```

```
        for(j = 0; j < 1000; j++);
```

```
}
```

```
void main()
```

```
{
```

```
    while(1)
```

```
    {
```

```
        LED1 = 1;
```

```
        LED2 = 0;
```

```
        delay();
```

```
        LED1 = 0;
```

```
        LED2 = 1;
```

```
        delay();
```

```
    }
```

```
}
```

OUTPUT

Registers

Register	Value
r0	0x00
r1	0x00
r2	0x00
r3	0x00
r4	0x00
r5	0x00
r6	0x00
r7	0x00
SP	0x07
PC	0x081D
STATUS	0x00000000
SEC	0.000389...
PWR	0x00

```
14: void main()
15: {
16:     while(1)
17:     {
18:         LED1 = 1;
19:     }
20: }
```

Command

Running with Code Size Limit: 2K
Load "C:\\Keil_v5\\CS1\\Examples\\HELLO\\Objects\\exp11"

Call Stack - Locals

Name	Location/Value	Type
MAIN	C:0x081D	

Simulation 11: 450.46049900 sec L:18 C:1 CAP: NUM SCRL OVR: R/W

Registers

Register	Value
r0	0x00
r1	0x00
r2	0x00
r3	0x00
r4	0x00
r5	0x00
r6	0x00
r7	0x00
SP	0x07
PC	0x081D
STATUS	0x00000000
SEC	0.000389...
PWR	0x00

```
14: void main()
15: {
16:     while(1)
17:     {
18:         LED1 = 1;
19:     }
20: }
```

Command

Running with Code Size Limit: 2K
Load "C:\\Keil_v5\\CS1\\Examples\\HELLO\\Objects\\exp11"

Call Stack - Locals

Name	Location/Value	Type
MAIN	C:0x081D	

Simulation 11: 709.1185900 sec L:18 C:1 CAP: NUM SCRL OVR: R/W